

David Shumway

Software Engineer | Machine Learning | Backend & Data Systems

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Summary

Senior Software Engineer and Machine Learning Engineer with 10+ years of experience building production backend systems and applied ML pipelines. Master's graduate (CS, UIC) specializing in knowledge graphs, domain adaptation, and transfer learning for environmental and biomedical data. Track record of end-to-end project ownership, from software design and machine learning model development and deployment, to distributed data pipelines, ETL/ELT orchestration, and monitoring, across both research and client-facing environments.

Employment History

Graduate Research Assistant (ML / Knowledge Graphs), University of Illinois at Chicago - Chicago, IL 2021-2026

- Lead developer on ONR-funded project (#00014-21-1-2286) applying ML to predict risk to Navy divers at coastal locations.
- Built and compared supervised, semi-supervised, and unsupervised models (LSTM, GRU, GNN, XGBoost, CNN, MLP, Random Forest) for time-series water quality prediction.
- Implemented domain adaptation and transfer learning methods (Feature Augmentation, Balanced Weighting, CORAL, subspace alignment) to generalize models to data-scarce coastal sites, published at IEEE BigData 2024, ES&T 2025.
- Designed and orchestrated ontology-driven ETL/ELT pipelines integrating environmental datasets (NOAA, EPA, WaterQualityData.us) for knowledge graph construction, published at ACM Web Conference 2023.
- Built OWL ontologies in Protégé for waterborne illness risk assessment; queried with SPARQL and RDF.
- Integrated large-scale public datasets (NOAA, EPA, NDBC, NREL, EN4) to improve coastal water quality ML models.

Graduate Research Assistant (Biomedical ML), University of Illinois at Chicago - Chicago, IL 2018-2021

- Built MALDI-DB (NIH #R01GM125943), a community-driven public repository with ML workflows for MALDI-TOF spectra, deployed via Docker/OpenStack on MassOpenCloud (Python, Django, R, PostgreSQL, nginx, D3.js).
- Developed CNN-based models for bacteria identification in MALDI-TOF mass spectrometry data.
- Co-authored grant proposal for EarthCube semantic data infrastructure connecting distributed scientific datasets.

Graduate Teaching Assistant, University of Illinois at Chicago - Chicago, IL 2018-2026

- Served as Instructor of Record for CS586: Data and Web Semantics, taking over mid-semester covering ontology engineering, RDF, OWL, and semantic web.
- Led discussions and mentored students across 12 semesters in courses including Big Data Mining, Software Design, Languages & Automata, and Mobile Development (groups of 5–250).
- Mentored 4 semesters of NSF-REU undergraduate researchers.

Senior Software Engineer, Scalesology - Remote 9/2024-2/2026

- Led backend development for a production CRM platform (insurance client): system design, PostgreSQL RDS on AWS, scalable API layer, and client-facing dashboards, maintaining a sustained contribution cadence of 150-250 commits per month over 7 months.
- Managed a suite of 10+ PowerBI dashboards, with data served via Snowflake, for an oil recycling client, containing high-level KPIs and operational metrics. Took over full administration and development when the client's lead analyst departed, adding new features, maintaining existing reports, and performing user access and security administration.
- Built and maintained Tableau reports for a financial services client, taking a handful of reporting templates and populating them with real client data including year-over-year earnings, quarterly performance metrics, and company financial KPIs. Built a custom Python ETL script to transform raw data from external reporting tools into Tableau-ready format.
- Built a PowerBI proof-of-concept for a grocery store client, creating a top-level overview of their

product dataset with financial and commerce-oriented metrics (product performance, inventory turnover, sales trends). Designed multiple pages with tables, charts, and a custom data matrix visualization, delivering actionable insights from millions of rows of grocery product data.

- Championed the adoption of AI development by leveraging tools such as GitHub Copilot daily for generating and reviewing PRs, code stubs, system design, and skeleton code, and use of autocomplete in VS Code, with careful review to ensure quality and integration.
- Built and maintained Tableau reports for a financial services client, taking 2-3 reporting templates and populating them with real client data, then updating them with latest data on a regular cadence to support ongoing performance tracking and decision-making.
- Migrated legacy SQL Server procedures to Python/Pandas services backed by Snowflake data warehousing workflows.
- Built interactive analytics dashboards and ETL/ELT data pipelines supporting client analytics workflows across energy, grocery, finance, and education sectors (ECharts, React, PowerBI, Tableau, D3.js).
- Implemented and utilized CloudWatch logging across multiple client cloud environments (AWS).
- Assisted with an internal LLM chatbot POC for document Q&A, running models locally on WSL to evaluate secure in-house document processing, with a focus on local setup and early integration work toward CSV, Google Drive, and SharePoint ingestion.
- Evolved building clustering tool running on AWS Lambda from using DBSCAN to using HDBSCAN in order to provide a confidence metric for clusters; and installed the Microsoft Building Footprints dataset (GlobalMLBuildingFootprints) via AWS PostGIS, integrating it with the Lambda tool as another heuristic by adding SQL queries to determine whether an address' given Latitude/Longitude coordinate fell within one of the dataset's building footprints.
- Used productivity tools daily toward cross-client collaboration and sharing documentation and deliverables (SharePoint); participating in daily standups and client meetings (Teams); internal team communication (Slack); managing client and internal code (GitHub); daily timesheets (QuickBooks); making progress on agile project deliverables (Jira); and WSL/Ubuntu and VS Code for starting dev servers, managing Git, running migrations, executing ETL, managing services, and developing and programming in a Windows environment.

Founder & Lead Developer, Finger Lakes Business Systems - Remote

2010-2018

- Collaborated with 750+ clients on Amazon MTurk, completing 3M+ microtasks across 1100+ unique projects with \$185K revenue while maintaining a 99.98% task approval rating, continuing off-site with key clients.
- Engineered in-house web scraping, data management, workflow optimization, and distributed systems solutions, writing 375K+ lines of in-house code in JavaScript, PHP, HTML/CSS, SQL, and Python, with extensive use of Google Chrome API and browser extension programming.
- Created and maintained a simple open-source browser extension to help streamline the MTurk worker interface, published to the Chrome Store and Firefox Add-Ons, reaching up to 10K weekly users.
- Designed factory-based browser extension architecture (~300K LOC JS) with tab-pipelined execution, preview-mode admission control, and cross-worker disk-cached scraping, achieving effective average hourly wages 3-6x above published worker benchmarks, with a MySQL backend storing 100s of millions of records (scraper data, task payloads, completion logs).
- Designed 300+ reusable library methods and 110+ factory methods (JavaScript), enabling rapid deployment of new task types by assembling pre-built components rather than writing custom code from scratch.
- Built 150+ PHP API endpoints for backend services, from file existence checks, disk-cached request utilities, server-side queueing with file locking, and SQL execution, to PDF-to-image conversion, and PDF-to-OCR, exposing numerous server-side capabilities to browser extensions.
- Engineered programmatic data extraction pipelines for financial and government data sources, including FINRA profiles, EDGAR filings, Cook County Clerk and Tax portals, NYC e-Services (nycserv.nyc.gov), physician profiles (npiregistry.cms.hhs.gov), and quarterly credit union call reports.
- Implemented strategies to handle CAPTCHAs and rate limiters, maintain logged-in sessions, and retrieve structured data from complex websites.
- Integrated external data into task workflows, with automated submission and lightweight worker validation, ensuring high data quality and 99.98% approval rates across thousands of tasks while achieving hourly wages 3–6x above published benchmarks.

Software Engineer, École Polytechnique Fédérale de Lausanne, DSIL - Lausanne, Switzerland

Summer 2017

- Joined Distributed Information Systems Laboratory for a summer internship at EPFL in Lausanne, Switzerland working alongside Martin Benjamin and Karl Aberer on the Kamusi project.

- Enhanced Kamusi.org's Neo4j natural language database by analyzing and integrating new datasets.
- Digital Archivist & Technical Specialist**, Paul Brunton Philosophic Foundation - Burdett, NY 2010-2014
- Curated 100K+ scanned documents and designed LAMP, JavaScript/HTML, Bash, and VBA solutions.
 - Automated scanning process, resulting in 4-8x efficiency and saving 2-4 years of manual work.
- Prior to 2010, held various roles including freelance web development (PHP/MySQL/JavaScript/HTML/CSS), audio transcription, food delivery, retail, and manual labor while self-teaching programming and building foundational technical skills.

Education

M.S. in Computer Science, University of Illinois at Chicago (August 2026) 2018-5/2026

- Advisors: Isabel Cruz, Cornelia Caragea.
- Completed PhD qualifying exam in *Advances in Scientific Workflow Management System Provenance Storage and Querying* (2020). 1.5 oral, 2.0 written (1-5, 1=best).
- Completed PhD coursework requirement (2021). Transitioned to M.S. program Summer 2026 with coursework-only option and awaiting official graduation confirmation (semester end).
- GRE (entrance): 159 verbal, 149 quantitative, 4.0 analytical.
- 41 GPA credit hours, 3.51/4.0 GPA. 104 PhD Thesis Research credit hours.
- Authored an 85-page PhD dissertation proposal framing the research agenda, including problem formulation, literature review, methodological approach, and preliminary results across domain adaptation, transfer learning, knowledge graph integration, scientific computing, and data-generating workflows and worker-built execution infrastructure in crowdsourcing environments.

B.S. in Computer Science, State University of New York at Plattsburgh 2015-2017

- Dean's List honors in all five semesters.
- 17th Annual Computer Science Department Academic Excellence Award recipient (2016).
- 97 GPA credit hours, 3.86/4.0 GPA.

No degree, Utah Valley University, Germanna Community College, Northeastern University 1998, 2001-2002

- Architecture, Arts and Sciences, & Engineering Focuses.
- 80 GPA credit hours.

Technical Skills

Languages: Python, JavaScript, PHP, HTML, CSS, Java, R, C, C++, C#, Swift, Assembly, VBA, VB.NET, Alice

Databases & Knowledge Graphs: MySQL, MariaDB, PostgreSQL, SQL Server, SQLite, Neo4j, Cypher, SPARQL, RDF, OWL, Protégé, WebProtégé, Virtuoso, RDFLib, pgAdmin, PhpMyAdmin, MySQL Workbench, ROBOT, CDC

Machine Learning, AI/LLM, Data Science: PyTorch, TensorFlow, Scikit-Learn, Pandas, GeoPandas, NumPy, PySpark, Dask, Jupyter, Google Colab, Matplotlib, domain adaptation, transfer learning, ADAPT, EasyAdapt, Balanced Weighting, CORAL, Subspace Alignment, GitHub Copilot, Agentic Workflows, Conversational AI, Prompt Engineering, RAG, LangGraph, LlamaIndex

Web Frameworks & Backend: Django, Flask, FastAPI, SQLAlchemy, Node.js, Next.js, serverless, REST, AJAX, Apache Cordova

Cloud & DevOps: AWS (RDS, S3, Lambda, ECS, EKS, CloudWatch, Route 53, IAM, Secrets Manager, Cognito), Azure (SQL, Key Vault, Fabric, Data Lake, OneLake), Docker, Docker Compose, Podman, OpenStack, Kubernetes, MassOpenCloud, CloudLab, n8n (production workflows, webhooks, ORM event integration), Apache Web Server, Nginx, VPS, GitHub Secrets

Web Scraping & Automation: Scrapy, cURL, HTTP/HTTPS, Browser Extensions, Greasemonkey, Userscripts, Chrome, Firefox, n8n, tesseract

Visualization: D3.js, Chart.js, Three.js, LaTeX, ECharts, matplotlib, PowerBI, Tableau, Leaflet

Frontend: JavaScript, TypeScript, HTML/CSS, Bootstrap, React, templating (Django/Jinja), custom JavaScript libraries

Dev Tools & Platforms: Git, GitHub, GitLab, Bash/Shell, Linux (Fedora, Ubuntu, Debian), Android, Chrome Extension API, SSH

Testing: JUnit, Pytest, Unittest, TDD

Productivity: Microsoft Teams, Slack, Jira, Discord, SharePoint, Git, GitHub, GitLab, GitHub Copilot, Claude, ChatGPT, VS Code, Gimp, Adobe Photoshop, Miro, Figma

HW/SW: Built and maintained PCs and laptops including Linux/Windows installation and maintenance.

Research Focus

Construction and use of knowledge graphs and ontologies, exploration of knowledge graph and ontology learning, and knowledge graph embeddings. Applying machine learning techniques in two key directions: 1) detecting bacteria species and small molecules in MALDI-TOF spectra, and 2) predicting concentrations of fecal indicator bacteria in coastal waters with transfer learning and domain adaptation.

Research Experience

Project Title: AI methods for reducing diver exposure to biological hazards

Research Advisors: Dr. Isabel Cruz, Dr. Cornelia Caragea

Duration: August 2021 - May 2026

Grant: ONR, #00014-21-1-2286

Project Title: A new paradigm for the creation and mining of microbial libraries for drug discovery

Research Advisor: Dr. Isabel Cruz

Duration: August 2018 - August 2021

Grant: NIH, #R01GM125943

Research Interests

Data science, web and data semantics, knowledge graphs, ontologies, embeddings, big data, machine learning, domain adaptation, transfer learning, databases, data mining, information retrieval, web search, information visualization, computational biology, open-source, agile development.

Teaching History

University of Illinois at Chicago - Chicago, IL

2018-2024

Instructor

Data and Web Semantics, CS586 (2021)

Teaching Assistant

Digital Literacy, CS100 (2019)

Languages and Automata, CS301 (2019, 2023, 2024)

Software Design, CS342 (2018)

Software Engineering I, CS440 (2019)

Software Development for Mobile Platforms, CS478 (2020, 2021)

Big Data Mining, CS494 (2020)

Data and Web Semantics, CS586 (2019, 2020, 2021)

Graduate Coursework – UIC (relevant)

Computer Algorithms, CS401

Introduction to Machine Learning, CS412

Visualization and Visual Analytics I, CS424

Systems Performance and Concurrent Computing, CS463

Numerical Analysis, MCS471

Software Development for Mobile Platforms, CS478

Computer Algorithms II, CS501

Data and Web Semantics, CS586

Research Methods, CS590

Algorithms for Big Data, CS594

Visual Data Science, CS594

Undergraduate Coursework – SUNY Plattsburgh (relevant)

Artificial Intelligence, CSC345

Operating Systems, CSC433

Data Mining, CSC442

Software Design Studio, CSC446

Reliable Systems, CSC456

Climate Change Science, ENV406

Sustainable Transportation, GEG380

Urban Geography and Planning, GEG307

Linear Algebra I, MAT202

Sets, Functions and Relations, MAT231

Manuscripts & Publications

† co-author

* primary author

‡ supporting role

Published

- * Shumway, D. (2026, July). Engineering a Quota-Constrained Human-in-the-Loop Data Platform: Marketplace-Aware Architecture over 3M+ Microtasks. Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7040318
- * Shumway, D. (2026, March). MALDI-DB: Architecture and Implementation of a Community-Driven Platform for Bacterial Identification via MALDI-TOF Mass Spectrometry. Technical report, University of Illinois at Chicago. Published online: <https://davidshumway.github.io/self/MALDI-DB>
- † Elahi, A., Shumway, D., Kowalczyk, M., Shrestha, A., Caragea, D., Caragea, C., & Dorevitch, S. (2025). Machine Learning, Generalization, and Transfer Learning for Predicting the Exceedance of Fecal Indicator Bacteria Thresholds at Beaches. *Environmental Science & Technology*, 59(42), 22386-22396.
- † Elahi, A., Shumway, D., Kowalczyk, M., Shrestha, A., Gautam, N., Caragea, D., Caragea, C., & Dorevitch, S. (2024, December). Predicting Surface Water Bacteria Levels Using Transfer Learning and Domain Adaptation. In *2024 IEEE International Conference on Big Data (BigData)* (pp. 1-10). IEEE.
- † Gautam, N., Shumway, D., Kowalczyk, M., Khanal, S., Caragea, D., Caragea, C., & Dorevitch, S. (2023, April). Leveraging existing literature on the web and deep neural models to build a knowledge graph focused on water quality and health risks. In *Proceedings of the ACM Web Conference 2023* (pp. 4161-4171).

Submitted & Revised

- † Submitted to *IJCAI (2023)*, *DSAA (2023)*, *The Web Conference (2024)*.
- * Shumway, D., Elahi, A., Gautam, N., Kowalczyk, M., Caragea, D., Caragea, C., & Dorevitch, S. (2023). A Domain Adaptation Approach for Predicting Recreational Water Quality at Data-Scarce Coastal Locations. Submitted to *Environmental Science & Technology*.

No Submission

- * Shumway, D., Elahi, A., Gautam, N., Kowalczyk, M., Caragea, D., Caragea, C., & Dorevitch, S. (2023). Developing a recreational and occupational waterborne illness knowledge graph toward biological risk assessment in coastal waters.
- * Shumway, D. (2023). Use of large-scale public datasets toward improving coastal water quality predictions.
- * Aggarwal, M., Shumway, D., Cruz, I. (2020). Bacteria Identification in MALDI-TOF Mass Spectrometry Data Using a Convolutional Neural Network Approach.
- * Shumway, D. (2020). Storage and Querying of Provenance in Scientific Workflow Management Systems. Completed as part of PhD qualifying exam (UIC).
- † Zhao, W., Shumway, D. (2020) A Review of QA4IE. Completed as part of Research Methods graduate course (UIC, CS590).
- * Shumway, D. (2020). A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry bacterial and protein spectral data and workflows. Completed as part of Research Methods (UIC, CS590).
- * Shumway, D. (2019). Bacterial identification in a distributed mass spectrometry sensor network using streaming big data. Completed as part of Algorithms for Big Data (UIC, CS594).
- † Zhao, K., Shumway, D., Xingbo, W., Cruz, I. (2019). Efficient Multi-user Semantic Desktop Framework that Supports Data-aware Storage.
- * Shumway, D. (2018). ADEPT - An ontology for Atomic Distributed Experiments: Modeling Scientific Workflow Processes as Always-Available Cloud Services to aid in Workflow Reproducibility, Composition, Sharing, Resource Conservation, and Testability. Completed as part of Data and Web Semantics (UIC, CS586).

Grant Writing

- ‡ Dorevitch, S., Cruz, I., Catlett, C. (2021). Microbial hazard risk estimation and communication for Navy divers.
- ‡ Cruz, I., Miranda, F., Derrible, S., Siciliano, M., Cailas, M., Sambanis, A., ... & McHenry, K. (2021). ICE—PRESUR: Illinois Center of Excellence—Planning a Resilient and Equitable State Using Real-time data.
- † Cruz, I., Shumway, D. (2021). EarthCube Capabilities: Semantic Data Connections to Enable Distributed Science and Capability Building.
- † Zhao, K., Shumway, D., Xingbo, W., Cruz, I. (2019). Efficient Knowledge Graph Processing in Modern Memory and Storage Hierarchy.

- * Shumway, D. (2019). A collaboration and visualization tool to help meet sustainability goals by individuals and groups within local and global contexts. Completed as part of Grant Writing (UIC, UPP493).

Presentations

A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry protein fingerprints and workflows. Spring 2020. Presentation for PhD Written Critique and Proposal (WCP) PhD Qualifying Exam, UIC. Spring 2020.

A public community-driven database for sharing and execution of matrix-assisted laser desorption ionization time-of-flight mass spectrometry protein fingerprints and workflows. Spring 2020. Final presentation for Research Methods, UIC.

Bacterial identification in a distributed mass spectrometry sensor network using streaming big data. Spring 2019. Final presentation for Algorithms for Big Data, UIC.

ADEPT Ontology: Atomic Scientific Workflow Processes as Always-Available Cloud Services. Fall 2018. Presentation for Data and Web Semantics, UIC.

Natural Drug Discovery Web-Based Data Visualization: Project Proposal. Fall 2018. Presentation for Visual Data Science, UIC.

The Salvation of Paris: Notes on Sustainable Transit in Paris from Taras Grescoe's Straphanger. December 2017. Final presentation for Sustainable Transportation, SUNY Plattsburgh.

Containers, Docker, Linux: Notes from the 2016 LinuxCon Conference. August 2016. Presented to the Software Engineering Club at SUNY Plattsburgh.

Health and Well-Being in the Built Environment: Notes on Christopher Alexander. May 2016. Final presentation for Public Speaking, SUNY Plattsburgh.

Open Source Contributions

- Published Tools for Amazon's Mechanical Turk (Chrome, Firefox) on GitHub, a public productivity browser extension for AMT. Included documentation, issue tracking, and bugfixes. Used by up to 10,000 users per week.
- Publicly shared numerous small projects and code snippets on GitHub and GitHub Gists demonstrating a commitment to community and collaboration in software development.
- Contributed to open-source issue tracking through new issue documentation and creation of pull requests.

Conference Attendance

International Conference of Biomedical Ontologies, Virtual attendee. (2022).

EarthCube Annual Meeting, Virtual attendee. (2021, 2022).

UIC Urban Forum. (2019).

LinuxCon, Toronto. (2016).

Notebook Reviewer

2nd and 3rd Annual EarthCube Call for Notebooks. (2021, 2022).